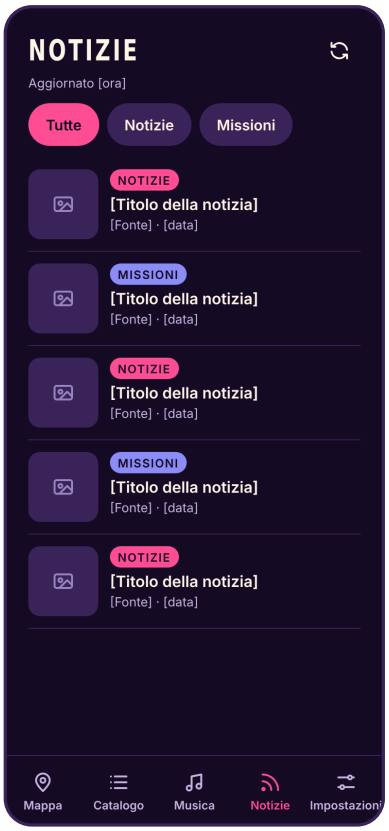
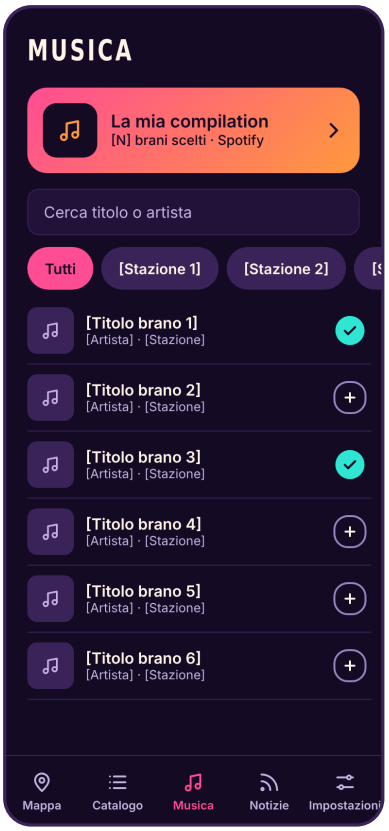
IloveVI

Functional Analysis

Version 0.4 (draft for review) | 18 October 2026
Unofficial companion app for mobile (Android and iOS)







Screen previews are rendered from the design mockups with substitute fonts and placeholder content (text in square brackets). Labels are in Italian, the language of the mockups; Appendix A gives the English equivalents.

Contents

1. Purpose and status

2. Product overview

3. Feature overview

4. Navigation and screen inventory

5. User flows

6. Functional requirements

7. Screen specifications

8. Data model

9. Integrations

10. Non-functional requirements

11. Design gaps

12. Assumptions and open questions

13. Delivery phases

14. Appendix A: UI label glossary

1. Purpose and status

This document describes what IloveVI does, screen by screen, so that design, development and testing share one reference. It is based on the eight screens designed so far and on the decisions taken during the design phase. Anything that is not visible in the designs, but is needed to make the app work, is marked with (*) and is a proposal to be confirmed.

Priorities use three levels: **Must** (needed for the first release), **Should** (important, can follow closely) and **Could** (nice to have). They are proposed, not agreed.

Status: draft. Open decisions are collected in section 12.

Revision history

Version	Changes
0.1	First draft.
0.2	The map no longer shows or centers on the user's position (none in this version). New News feature: stories about the game and its missions, read from RSS feeds, with a News screen and a fifth bottom-bar destination.
0.3	Map source changed: the map is a remote image that the project owns or may use, cut into tiles by a pipeline, instead of an original illustration. New internal placement tool to position pins on the map. New MapPin and MapConfig entities, requirements MST and PLT, and a new delivery order.
0.4	The "health" item type is removed (it does not exist in GTA VI); item sub-types are now only placeholders (ammo, armor) pending the real classification. Pin clustering is now a Must: when more than 10 pins sit too close together on screen they merge into a round count badge (requirement MAP-10, section 7.1). New icon set: one icon per sub-type (pistols for weapons, car, plane and boat for vehicles, box for items), 13 icons in total (requirement MAP-02).

2. Product overview

IloveVI is an unofficial companion app for players of Grand Theft Auto VI ("the game"). It is used on a phone next to the game and answers four questions quickly: where a given weapon, vehicle or item is on the map, what it is, which songs are on the in-game radio, and what is new about the game and its missions. It also lets the player turn a selection of songs into a Spotify playlist to listen to while playing.

Target user and context

- One type of user: a player with the game running on a console or PC and a phone within reach.
- Short, frequent look-ups during a pause, often in low light and with one hand.
- Connectivity may be poor, so the app must work offline.

Design principles

- **Read-only utility.** The app does not track missions, progress or completion. The user looks, finds, and goes back to the game.
- **Offline first.** All content, news included, is stored on the device and updated in the background.
- **Three colors, three categories.** Pink is always Weapons, orange Vehicles, teal Items, on the map and in every list.
- **One theme, dark,** with large touch targets (44 px minimum).

In scope

- Interactive map with pins and filters
- Catalog and detail of weapons, vehicles and items
- In-game song library
- Song compilation and Spotify playlist creation
- News feed about the game and its missions, read from RSS feeds
- Settings, content updates and legal disclaimer
- Internal placement tool (not part of the app) to position pins on the map

Out of scope

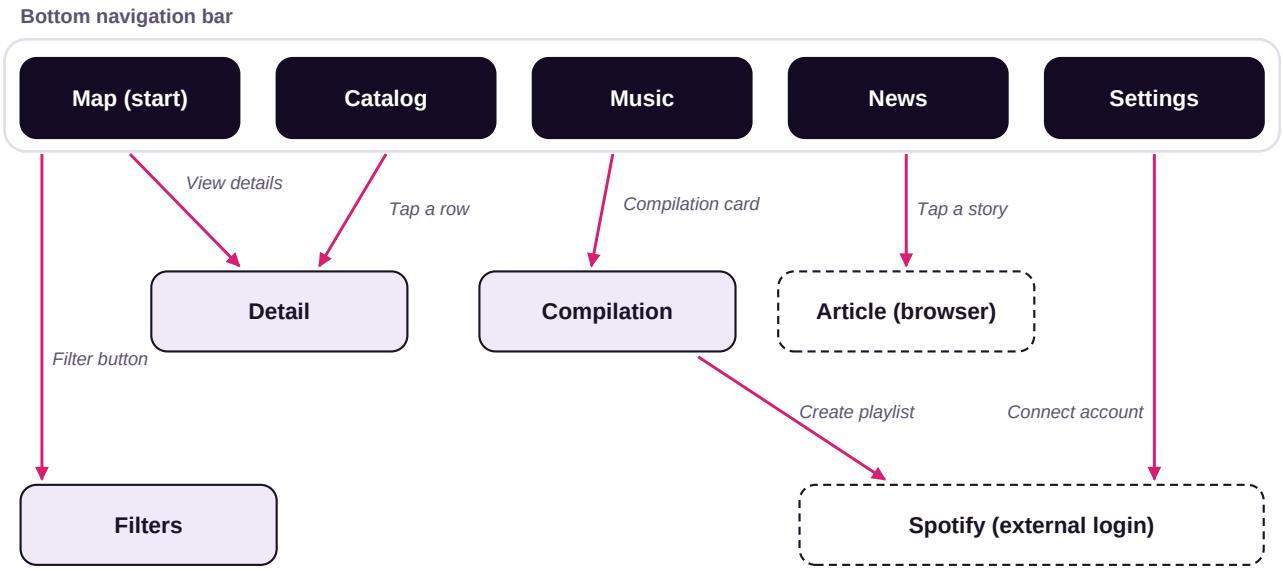
- Missions as a game feature: checklists, completion percentages or any progress tracking (removed on purpose). Missions appear only as a topic of news stories.
- The player's position: the map shows no user location and the app requests no location permission (not in this version).
- Reproducing news articles: stories open at their source, the app shows only headline, short excerpt and thumbnail.
- User accounts, cloud sync of user data, community or crew features.
- Any integration with the game itself: the app cannot read the game state, player position or audio.
- Playing music inside the app, lyrics, audio files, and any official game asset (logos, fonts, artwork, maps).
- Monetization: not defined at this stage.

3. Feature overview

ID	Feature	What it does	Screens	Priority
F1	Interactive map	Pan and zoom map with colored pins for weapons, vehicles and items; filter by category and sub-type.	Map, Filters	Must
F2	Catalog	Browse and search every weapon, vehicle and item.	Catalog	Must
F3	Entry detail	Image, location mini-map, attributes, and a jump to the map.	Detail	Must
F4	Song library	All in-game songs, with station filter and search.	Music	Must
F5	Compilation	Pick and order songs; one compilation kept on the device.	Music, Compilation	Must
F6	Spotify playlist	Create a playlist of the compilation in the user's Spotify account.	Compilation, Settings	Must (see risk, 9.1)
F7	News feed	Latest stories about the game and its missions from RSS feeds; filter by topic; opens the original article.	News	Must
F8	Offline content updates	Works offline; downloads new content without an app release.	Settings	Must
F9	Settings	Spotify account, map labels, data update date, language, disclaimer.	Settings	Should
F10	Pin placement tool (internal)	Desktop tool to place weapons, vehicles and items on the map, edit and import pins, and publish them to the backend.	Internal tool	Must

4. Navigation and screen inventory

The app has five top-level destinations in a bottom bar. Secondary screens open on top of them and close with a back control. Compilation is reached only from Music and is not in the bar.



Arrows show forward navigation; dashed boxes are outside the app. Every secondary screen has a back control, and "Show on map" in Detail returns to the Map with the entry selected.

#	Screen	Design file	Reached from	Leads to
1	Map	Main.dc.html	App start; bottom bar; Detail ("Show on map")	Filters, Detail, map search (*)
2	Filters	Mappa.dc.html	Map filter button	Map
3	Catalog	Categoria.dc.html	Bottom bar	Detail
4	Detail	Dettaglio.dc.html	Map summary card; Catalog row	Map
5	Music	Musica.dc.html	Bottom bar	Compilation
6	Compilation	Compilation.dc.html	Music compilation card	Music, Spotify
7	News	News.dc.html	Bottom bar	Article in the browser
8	Settings	Impostazioni.dc.html	Bottom bar	Spotify

5. User flows

F-A. Find a weapon on the map

1. Open the app: the Map is shown with all three categories on.
2. Optionally switch category chips on or off, or open Filters and choose sub-types, then tap "Show on map".
3. Tap a pin: it grows with a ring and a summary card appears with name, category and zone.
4. Tap "View details" to open Detail, then "Show on map" to return to the Map centered on the entry.

F-B. Look a vehicle up by name

1. Open Catalog and tap the Vehicles chip.
2. Type part of the name or the zone in the search field: the list filters as you type.
3. Tap a row to open Detail, then "Show on map" to see where it is.

F-C. Build a compilation

1. Open Music and, if needed, pick a station chip or search by title or artist.
2. Tap "+" on a song: it turns into a teal check and the compilation card count increases.
3. Open the compilation card to review the list, drag songs to reorder and tap X to remove.

F-D. Create the Spotify playlist

1. In Compilation tap "Create the playlist on Spotify".
2. If Spotify is not connected, the user is asked to connect and signs in on Spotify's own page.
3. The app matches each song to a Spotify track and creates the playlist in the user's account.
4. The result is shown (created, partially matched, or failed); the user opens Spotify and plays it while gaming.

F-E. Receive new content

1. At start, and whenever online, the app checks for content changes and downloads only what changed.
2. Settings shows the date of the last update; the user keeps using the app while it updates.

F-F. Read the news

1. Open News: the latest stories from all feeds are listed, newest first, with the time of the last refresh.
2. Optionally tap the Notizie (News) or Missioni (Missions) chip to see only that topic.
3. Tap the refresh icon to check the feeds again; if offline, the stored stories stay visible.
4. Tap a story to open the original article in the phone's browser.

6. Functional requirements

Rows marked (*) are not shown in the current screens and are proposed additions.

6.1 Navigation

ID	Requirement	Priority
NAV-01	Bottom bar with five destinations: Map, Catalog, Music, News, Settings. The active one is highlighted.	Must
NAV-02	The Map is the start screen.	Must
NAV-03	Filters and Detail have a back control that returns to the previous screen.	Must
NAV-04	Compilation opens only from the compilation card in Music and is not in the bottom bar.	Must

6.2 Map

ID	Requirement	Priority
MAP-01	Show the game map as an image the user can pan and zoom (pinch, double tap, + and - buttons).	Must
MAP-02	Show one pin per located entry, colored by category: Weapons pink, Vehicles orange, Items teal. The icon follows the sub-type: pistol, rifle, machine gun, blade or bomb for weapons; car, motorcycle, boat, helicopter or plane for vehicles; ammo, armor or a generic box for items.	Must
MAP-03	Three category chips show or hide each category. All are on at first launch.	Must
MAP-04	A filter button opens Filters. "Show on map" applies the choice and returns; "Reset" turns every sub-type back on.	Should
MAP-05	Tapping a pin selects it (larger, with a ring) and opens a summary card: icon, name, "category - zone" and a "View details" button. Only one pin is selected at a time.	Must
MAP-06	Tapping the map background clears the selection and closes the card. (*)	Should
MAP-07	The search icon opens a search by name or zone; picking a result centers the map and selects the pin. (*)	Could
MAP-08	The map never shows the user's position: there is no position marker and no center-on-player control, and the app does not ask for location permission. Position may be reconsidered in a later version.	Must
MAP-09	Pins can show the entry name as a label, controlled by a Settings switch.	Could
MAP-10	Pins that are too close on screen are grouped into a cluster so the map stays readable. Rule: pins closer than 48 px to each other form a group; when a group has more than 10 pins it is drawn as ONE round badge showing the number of pins (99+ above 99); a group of 10 or fewer stays as individual pins. The badge takes the color of the dominant category, or the brand color when mixed. Tapping a badge zooms in on the group; as the user zooms, groups split and badges shrink to individual pins. At maximum zoom no badge is drawn. Category and sub-type filters apply before grouping, so the number counts only visible pins. The threshold (10) and distance (48 px) are single constants that can be changed.	Must
MAP-11	The selected pin is never hidden inside a badge: while a pin is selected (for example after "Show on map"), it stays drawn on top and its group is split around it.	Should

6.3 Catalog and Detail

ID	Requirement	Priority
CAT-01	List the entries of the selected category (Weapons first) with thumbnail, name, "type - zone" and a chevron.	Must
CAT-02	Three chips (Weapons, Vehicles, Items) switch category; exactly one is active.	Must
CAT-03	A search field filters the list by name or zone as the user types.	Must
CAT-04	Tapping a row opens Detail.	Must
CAT-05	When nothing matches, show an empty state with a hint. (*)	Should
DET-01	Show category badge, name, type and image.	Must
DET-02	"Where to find it": a mini-map with the pin, the zone name and a short description.	Must
DET-03	"Show on map" opens the Map centered on the entry with its pin selected.	Must
DET-04	Show an attributes table. Weapons: type, price, ammo. Vehicles and items need their own sets. (*)	Should

6.4 Music and Compilation

ID	Requirement	Priority
MUS-01	List the in-game songs with thumbnail, title and "artist - station".	Must
MUS-02	Search by title or artist.	Must
MUS-03	Station filter: an "All" chip plus one chip per station, scrolling horizontally, one active at a time.	Must
MUS-04	The "+" control adds a song to the compilation; added songs show a teal check and the control removes them.	Must
MUS-05	A compilation card at the top shows name, song count and the Spotify label, and opens Compilation.	Must
MUS-06	Each song offers an "Open in Spotify" link that needs no login. (*)	Could
CMP-01	Show cover, name, song count and total duration.	Must
CMP-02	List the songs in order with a handle to reorder and an X to remove.	Must
CMP-03	"Add more songs" returns to Music.	Must
CMP-04	"Create the playlist on Spotify" needs a connected account; if none, ask to connect first. Then match every song to a Spotify track and create the playlist in the user's account.	Must
CMP-05	Show the outcome: created (open in Spotify), partially matched (N of M found, with the missing songs listed), or failed with retry. (*)	Must
CMP-06	Under the button, state that Spotify cannot be linked to the game and the playlist is played separately.	Must
CMP-07	The compilation is stored on the device. The first release has a single compilation.	Must
CMP-08	The user can rename the compilation. (*)	Could

6.5 News

ID	Requirement	Priority
NWS-01	A News screen lists stories read from a set of RSS or Atom feeds, newest first, each with thumbnail (if the feed has one), topic badge, headline, and "source - date".	Must
NWS-02	Every feed is tagged with a topic, News or Missions. Topic chips (All, News, Missions) filter the list; All is active at first.	Must
NWS-03	Tapping a story opens the original article in the phone's browser. The app shows only headline, a short excerpt, thumbnail and the source name, never the full article.	Must
NWS-04	A refresh icon fetches the feeds again. The screen shows "Updated [time]" for the last successful refresh.	Must
NWS-05	Feeds are also refreshed in the background (at app start and at most every [30] minutes while the app is open). (*)	Should
NWS-06	Stories are stored on the device so the list shows offline. Only the latest [200] stories are kept. (*)	Must
NWS-07	The list of feeds (address, name, topic, on/off) is part of the backend content, so feeds can change without an app release.	Must
NWS-08	Stories from several feeds are merged and duplicates (same link or same headline) are shown once. (*)	Should
NWS-09	If one feed fails, the others still load and the failure does not block the screen. If all fail, the stored stories stay visible with an error message and a retry action. (*)	Must
NWS-10	Stories with no image show a neutral placeholder; images load only when needed. (*)	Should

6.6 Settings

ID	Requirement	Priority
SET-01	Spotify card: connect and disconnect the account, with the connection status. (*) for disconnect and status.	Must
SET-02	Switch "Show labels": names next to map pins.	Could
SET-03	Show the date of the last content update, with a manual refresh. (*) for refresh.	Should
SET-04	Theme row: the app has a single dark theme, so the row is informational.	Could
SET-05	Language selection: Italian and English. (*)	Should
SET-06	Always show the disclaimer: unofficial app, not affiliated with Rockstar Games or Take-Two Interactive.	Must

6.7 Content and data

ID	Requirement	Priority
DAT-01	The app works fully offline with the content bundled in the build and the last downloaded update.	Must
DAT-02	Content (entries, pins, map configuration, songs, stations, news feed list) comes from a read-only backend and only changes since the last update are downloaded.	Must
DAT-03	Content updates never require a new app release.	Must
DAT-04	No account is needed. No personal data is collected except what the optional Spotify connection requires. No location data is used.	Must
DAT-05	Preferences, the compilation and the news cache are stored locally.	Must

6.8 Map source and tiles

ID	Requirement	Priority
MST-01	The map is one source image stored remotely. The project must own or hold the right to use it; the official game map and fan-made maps are not used without permission.	Must
MST-02	A backend record (MapConfig) gives the image address, width and height in pixels, map version, tile address template, minimum and maximum zoom.	Must
MST-03	A pipeline cuts the source image into tiles at several zoom levels and stores them on the backend storage or a CDN. The app never loads the full source image.	Must
MST-04	The app loads tiles on demand and keeps them in a disk cache. The lowest zoom level is downloaded at first launch so the map is never blank offline. (*)	Must
MST-05	Pin positions are normalized (0 to 1) against the source image, so they do not depend on screen size, tile size or zoom.	Must
MST-06	Every pin records the map version it was placed on. If the map version changes, pins from older versions are flagged for repositioning. (*)	Should
MST-07	While a tile is loading, show a neutral placeholder; if a tile fails, keep the rest of the map usable. (*)	Should

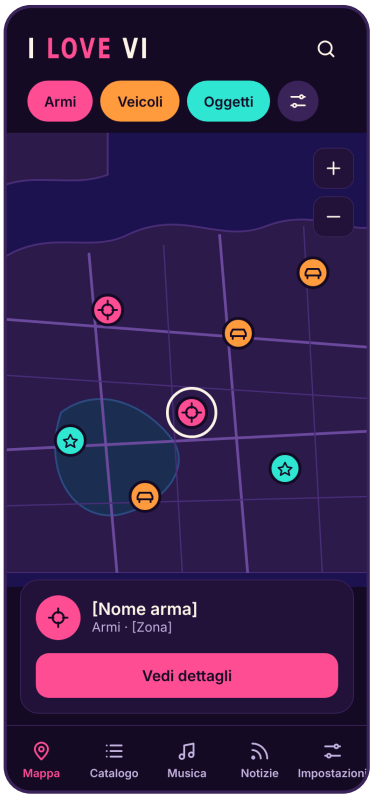
6.9 Pin placement tool (internal)

ID	Requirement	Priority
PLT-01	The tool is a desktop application of the project, not part of the mobile app, and requires an administrator sign-in.	Must
PLT-02	It shows the map with pan and zoom, using the same tiles and the same normalized coordinates as the app.	Must
PLT-03	A click on the map creates a pin and shows its normalized coordinates. The pin is linked to an existing weapon, vehicle or item chosen from a searchable list.	Must
PLT-04	A pin can be moved by dragging, or by arrow keys for fine adjustment, and deleted.	Must
PLT-05	A list of all pins can be filtered by category and by entries that have no pin yet.	Should
PLT-06	Pins can be imported from and exported to CSV or JSON, so hundreds of records can be entered quickly after the game release.	Must
PLT-07	Saving writes the pins to the backend with an update date, so the app receives them with the normal content update and no app release.	Must
PLT-08	The tool can generate and publish the tiles of a new source image and register a new map version. (*)	Should

7. Screen specifications

Each page shows the designed screen and lists what it contains, how it behaves and what is still undefined. Square brackets in the previews are placeholder content that will come from the backend.

7.1 Map



The start screen. It lets the player locate weapons, vehicles and items on the game map.

Elements

- App wordmark and a search icon in the header.
- Category chips (Weapons, Vehicles, Items) and a filter button.
- Map with colored category pins; the selected pin is larger with a ring.
- Zoom in and zoom out buttons. No user position marker and no center-on-player button.
- Summary card: category icon, name, "category - zone", "View details" button.
- Bottom bar with Map active.

Behavior

MAP-01 to MAP-11, MST-04, MST-05, NAV-01, NAV-02.

Data shown

Pins (category, linked entry, normalized x/y, zone) and the entry name and category for the summary card; map tiles.

Not yet designed

- Map loading and error states, tile placeholder.
- All chips off (no pins visible).
- Card hidden when nothing is selected.
- Search results; cluster badge appearance (see section 11).

7.2 Filters



Refines which sub-types of pins appear on the map.

Elements

- Back control and title.
- Three groups of chips, colored by category: Weapons (pistols, rifles, machine guns, melee, explosives), Vehicles (cars, motorcycles, boats, helicopters, planes), Items (ammo, armor).
- Footer with "Reset" and "Show on map".

Behavior

MAP-04. All sub-types start selected; tapping a chip toggles it. "Show on map" applies and returns to the Map.

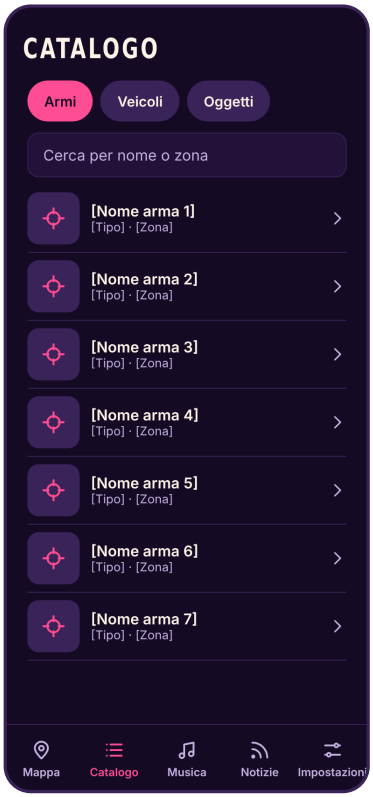
Data shown

The sub-types shown are generic placeholders (the item types in particular); they will be replaced by the real categories of the game.

Not yet designed

- Inactive chip appearance in the Filters screen.
- Behavior when every chip is off.

7.3 Catalog



A searchable list of every weapon, vehicle and item.

Elements

- Title and three category chips, one active.
- Search field "Search by name or area".
- Rows with thumbnail, name, "type - zone" and chevron.
- Bottom bar with Catalog active.

Behavior

CAT-01 to CAT-05. Tapping a row opens Detail.

Data shown

Name, type, zone, thumbnail for each entry of the active category.

Not yet designed

- Empty state when nothing matches.
- List loading state.
- Thumbnails for vehicles and items.

7.4 Detail



Everything about one entry, with its location. The design shows the weapon variant.

Elements

- Back control, category badge, name and type.
- Image area.
- "Where to find it": mini-map with the pin, zone and description.
- "Show on map" button.
- Attributes table: type, price, ammo.

Behavior

DET-01 to DET-04. "Show on map" returns to the Map with the entry selected.

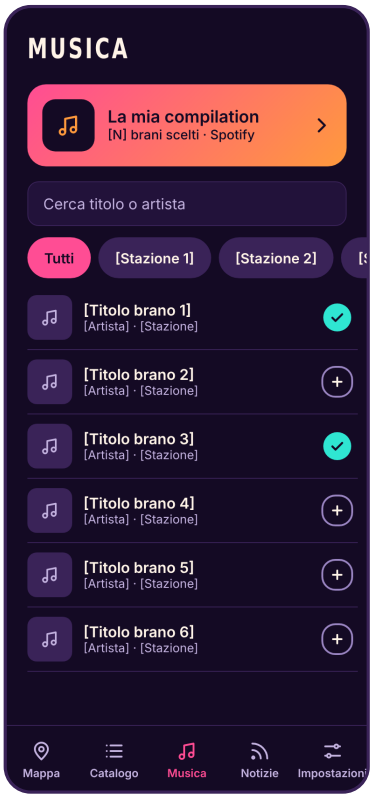
Data shown

Image, description, location text, coordinates and an attribute set that depends on the category.

Not yet designed

- Vehicle and item variants (different attributes).
- Missing image placeholder.
- Entries with no known location.

7.5 Music



The library of songs in the game, used to build a compilation.

Elements

- Title and the compilation card (name, song count, Spotify label).
- Search field "Search title or artist".
- Station chips: "All" plus one per station, scrolling sideways.
- Song rows: thumbnail, title, "artist - station", and a "+" or teal check.
- Bottom bar with Music active.

Behavior

MUS-01 to MUS-06, NAV-04.

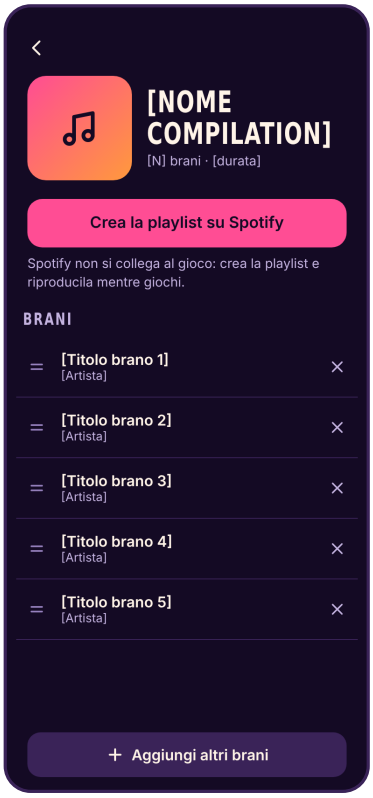
Data shown

Songs with title, artist and station; stations with name.

Not yet designed

- Empty state for no results.
- Compilation card when no song has been added.
- Optional "where it plays in the game" information (open question 3).

7.6 Compilation



The user's selection of songs, ready to become a Spotify playlist.

Elements

- Back control, cover, name, song count and duration.
- "Create the playlist on Spotify" button.
- Note that Spotify cannot be linked to the game.
- Ordered song list with reorder handle and remove button.
- "Add more songs" button.

Behavior

CMP-01 to CMP-08. The song list is stored on the device; the playlist is created through the Spotify integration (section 9.1).

Data shown

Compilation (name, ordered song ids, optional Spotify playlist id) and the songs it contains.

Not yet designed

- Empty compilation.
- Connect-to-Spotify prompt.
- Progress while matching and creating.
- Success, partial and failure results.
- Renaming.

7.7 News



The latest stories about the game and its missions, read from RSS feeds.

Elements

- Title and a refresh icon.
- "Updated [time]" line (Aggiornato).
- Topic chips: All, News, Missions (Tutte, Notizie, Missioni); one active.
- Story rows: thumbnail, topic badge, headline and "source - date".
- Bottom bar with News active.

Behavior

NWS-01 to NWS-10. Tapping a story opens the original article in the browser; the refresh icon fetches the feeds again.

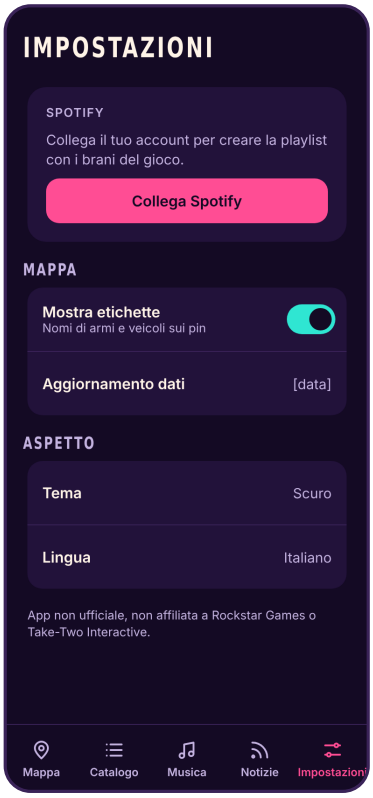
Data shown

Feed sources (name, address, topic) and cached stories (headline, link, date, short excerpt, image, source).

Not yet designed

- Loading, empty, offline and error states.
- Story without image.
- Pull-to-refresh and refresh progress.
- Whether a story shows its excerpt in the list.

7.8 Settings



Account connection and a few preferences.

Elements

- Spotify card with "Connect Spotify".
- Map section: "Show labels" switch and "Data update" date.
- Appearance section: theme (dark) and language.
- Disclaimer text.
- Bottom bar with Settings active.

Behavior

SET-01 to SET-06.

Data shown

Preferences stored locally; Spotify connection state; date of the last content update.

Not yet designed

- Connected state of the Spotify card.
- Manual refresh and its progress.
- Language picker.

8. Data model

Logical model, independent of the technology. Coordinates are normalized (0 to 1) so they do not depend on the resolution of the map image.

Entity	Main fields	Notes
Entry	id, category (weapon, vehicle, item), sub-type, name, description, zone, image, attributes, updatedAt	One record per weapon, vehicle or item. It has no coordinates of its own.
MapPin	id, category, entryId, x, y (normalized 0 to 1), zone, mapVersion, updatedAt	Position of an entry on the map. An entry with no pin does not appear on the map.
MapConfig	mapVersion, sourceImageUrl, widthPx, heightPx, tileUrlTemplate, minZoom, maxZoom, updatedAt	Describes the current map and its tiles.
Station	id, name, updatedAt	Radio station that plays a song.
Song	id, title, artist, stationId, spotifyTrackId (optional cache), updatedAt	No lyrics and no audio are stored.
FeedSource	id, name, url, topic (news or missions), enabled, updatedAt	A news feed to read. Part of the backend content.
NewsItem (local cache)	id (feed guid or link), feedId, title, link, publishedAt, excerpt, imageUrl, topic, fetchedAt	Cached on the device. Full article text is not stored.
Compilation (local)	id, name, ordered song ids, createdAt, spotifyPlaylistId (optional)	Stored on the device. One in the first release.
Preferences (local)	showLabels, language, last content update	Stored on the device.
Spotify session (local)	access token, refresh token, expiry	Stored in the platform secure storage.

9. Integrations

9.1 Spotify

- Purpose: create a playlist from the compilation in the user's own Spotify account.
- Sign-in uses the standard authorization flow with PKCE, opened in the system browser. The app needs permission to create and modify the user's playlists, and holds no secret.
- For each song the app searches Spotify by title and artist, keeps the best match and caches its track id.
- Songs with no match are reported to the user rather than skipped silently.
- Limit: Spotify cannot play inside the game. The app tells the user to play the playlist separately.
- Risk: the Spotify developer program limits the number of users of an app in development mode, and the rules for wider access change over time. The current conditions must be checked before this feature is promised for launch. Fallback: per-song links and a shareable list.

9.2 Content backend

- A hosted database holds entries, songs and stations. It is public and read-only: no sign-in is needed.
- Each record carries an update timestamp so the app downloads only the changes since its last update.
- Content entry is the main operational task after launch (hundreds of records in a few days), so an import from spreadsheet files and quick correction of single rows are needed.

9.3 News feeds (RSS)

- Purpose: show recent stories about the game and its missions without writing any editorial content.
- Format: RSS 2.0 and Atom feeds over HTTPS. The app reads title, link, date, short summary, and the image given by the feed (enclosure or media tag).
- Fetching: option A, the app downloads each feed directly (simple, no server cost, but some sites block apps or need CORS-free access and the app repeats the work on every device); option B, a backend job downloads and merges the feeds, and the app reads one clean list (more control, de-duplication, caching, and a single place to change sources). Proposed: B, with A as fallback. To confirm.
- Content rights: show only headline, short excerpt, thumbnail and source name, always link to the original article, and respect each site's terms of use. Which feeds to use must be checked against those terms.
- Sources: the feed list and the topic of each (News or Missions) are not defined yet (open question 11).

9.4 Map source and tiles

- The map is one source image stored remotely. The project must own the image or hold the right to use it. The official game map and fan-made maps are protected and are not used without permission.
- The image is downloaded once by the placement tool, cut into tiles at several zoom levels and published to the backend storage or a CDN. The app reads only tiles, never the full image, because a very large image does not fit in phone memory.
- If the source address changes or disappears, the app is not affected: it uses the published tiles.
- A map version number ties pins to the image they were placed on. Replacing or cropping the image creates a new version and the existing pins must be repositioned.
- The real game map can only be matched after release; until then a placeholder image is used.

9.5 Pin placement tool

- A desktop application written with Compose Multiplatform in the same repository, reusing the shared models (MapPin, normalized coordinates) and the map viewer of the app.
- Administrator sign-in on the backend: the public app reads only; the tool is the only writer of pins.
- Typical use: choose an entry from the list, click its place on the map, adjust, save. Bulk entry by CSV or JSON import.
- It is an internal tool with no visual design; it follows the design system tokens where convenient.

10. Non-functional requirements

Area	Requirement
Platforms	Android and iOS, built with Kotlin Multiplatform and a shared Compose interface.
Offline	Every screen works without a connection with the stored content; only Spotify actions, news refresh and opening articles need the network.
Performance	Smooth pan and zoom with several hundred pins (target to confirm), clustering recomputed in under 16 ms for 500 pins, tiles loaded on demand and cached on disk; fast list scrolling and search.
Accessibility	Touch targets of at least 44 px; text contrast as defined in the design system; text labels for every icon-only control.
Localization	Interface in English and Italian; all strings kept outside the code.
Security and privacy	Spotify tokens stored in the Android Keystore and iOS Keychain; no personal data collected; no location data and no location permission; privacy policy required by the stores.
Legal	Visible disclaimer; no official logos, fonts, artwork, maps, lyrics or audio; the app name is still under review for store acceptance.
Maintainability	Content from the backend, visual values from design tokens, components from the shared design system.
Monitoring	Crash reporting. Usage analytics are not decided.

11. Design gaps

States and screens needed to build the app that are not designed yet.

Area	Missing design
All lists and the map	Loading, empty and error states; offline indicator.
First launch	What the user sees before the first content download.
Map	Selection dismissal, search results, behavior with all categories off, tile loading placeholder.
Cluster badge	Not designed yet: round badge with the count (sizes for 11 to 99 and 99+), colors for single-category and mixed groups, selected state, tap feedback. Proposed: a circle of 40 px with a 2 px border in the dominant category color, number in the display font.
Placement tool	Not designed (internal tool). A functional layout is enough: map, pin list, entry picker, import and export.
Detail	Vehicle and item variants; entry without image or location.
Music and Compilation	Empty compilation, renaming, progress and result of playlist creation, Spotify error cases.
News	Loading, empty, offline and error states; story without image; refresh progress; row with excerpt.
Settings	Connected Spotify state, language picker, manual refresh progress.
Filters	Appearance of an inactive chip.

12. Assumptions and open questions

#	Topic	Question and proposal
1	Map image	Which remote image is the source, and does the project hold the right to use it? The official map and fan-made maps are protected. Proposed: use an image you own or have a license for, publish it as tiles, and keep a placeholder until then.
2	Spotify access	Can playlist creation ship at launch under the current developer-program limits? Proposed: ship per-song links first and add playlist creation when access is confirmed.
3	Where a song plays	Should a song also say where it is heard in the game (station only, or scene or place)? Proposed: station only.
4	Vehicle and item attributes	Which attributes does Detail show for vehicles and items?
5	Map search	What does the header search icon search, and how are results shown? Proposed: entries by name or zone, with results in a list.
6	Content entry	Who enters the data after launch and with which tool? Proposed: spreadsheet import into the backend.
7	Number of compilations	One or several? Proposed: one in the first release.
8	Content language	Are names kept as in the game or translated? Proposed: names as in the game; only the interface is translated.
9	App name	"IloveVI" refers closely to the game and may be rejected by the stores. A neutral name remains the safest option.
10	News language	You wrote "GTA V". This document assumes you mean news about GTA VI and its missions. Please confirm, or say whether GTA V stories are also wanted.
11	News sources	Which RSS feeds, and which of them are tagged News or Missions? Proposed: a few well-known gaming sites plus the publisher's own news feed, after checking their terms. Where a site offers no feed per topic, missions stories can be picked by keyword.
12	News fetching	Does the app read feeds directly or through the backend? Proposed: through the backend (section 9.3).
13	Item categories	Which item categories really exist in GTA VI? Health was removed because it does not exist; ammo and armor are still unconfirmed guesses. Until the game is out the item sub-types stay placeholders.
14	Cluster threshold	Clusters appear only for groups of MORE than 10 pins closer than 48 px, as requested. With smaller groups, several pins can still overlap a little at low zoom. Proposed: keep the rule as is and review it once real data is in; the threshold is a single constant.
15	Theme row	Keep an informational "Theme: Dark" row or remove it? Proposed: remove until a second theme exists.

Real content (positions, attributes, song list) can only be collected after the game is released, scheduled for 19 November 2026 at the time of writing. Until then the app is built and tested with placeholder data.

13. Delivery phases

Phase	Content	Result
1	Navigation, Catalog and Detail with bundled placeholder data	Browsable app, end to end
2	Music list, search, station filter, compilation building, per-song links	Useful without any login
3	Backend schema, map source and tile pipeline, pin placement tool	Pins can be placed and published
4	Map: tiled viewer, pan and zoom, pins, category chips, Filters, summary card, link to Detail	Core feature complete
5	Content updates (delta sync) and News feed (feeds, cache, News screen)	Content and news change without app releases
6	Spotify connection and playlist creation; Settings completed	Compilation becomes a playlist
7	Empty and error states, accessibility, localization, store submission	Release candidate

Appendix A: UI label glossary

The mockups are in Italian. This table gives the English meaning of every label in the previews.

Italian	English	Italian	English
Mappa	Map	Mostra sulla mappa	Show on map
Notizie	News	Vedi dettagli	View details
Missioni	Missions	Cerca per nome o zona	Search by name or area
Filtri	Filters	Dove trovarla	Where to find it
Catalogo	Catalog	Tipo / Prezzo	Type / Price
Musica	Music	Cerca titolo o artista	Search title or artist
Impostazioni	Settings	Tutti	All
Armi	Weapons	Stazione	Station
Veicoli	Vehicles	Aggiornato [ora]	Updated [time]
Oggetti	Items	Aggiorna	Refresh
Pistole	Pistols	Fonte	Source
Fucili	Rifles	[Titolo della notizia]	[Story headline]
Mitragliatrici	Machine guns	Brani	Songs
Armi bianche	Melee weapons	La mia compilation	My compilation
Esplosivi	Explosives	Crea la playlist su Spotify	Create the playlist on Spotify
Auto	Cars	Aggiungi altri brani	Add more songs
Moto	Motorcycles	Collega Spotify	Connect Spotify
Barche	Boats	Mostra etichette	Show labels
Elicotteri	Helicopters	Aggiornamento dati	Data update
Aerei	Planes	Aspetto	Appearance
Munizioni	Ammo	Tema / Scuro	Theme / Dark
Armatura	Armor	Lingua / Italiano	Language / Italian
Reimposta	Reset		